LPV Nonlinear QP-MPC

t_s: real current time
x_s: real current state
x_0: QP-MPC initial state
A_k: A-matrix at state k
B_k: B-matrix at state k
N: number of horizon time steps
U: stacked control vector
X: stacked state vector
S_x: State influences vector
S_u: Control influences matrix

MPC is solved at real time step t_s, state x_s, with initial state $x_0 = x_s$. MPC solves for optimal control vector U for N steps ahead. We use a quadratic program (QP) formulation within the MPC using a linear parameter-varying model.

We have the state space representation of the LPV as

$$x_{k+1} = A_k x_k + B_k u_k$$

where $k+1$ is the next time step, and the time varying matrices A_k and B_k are of the current time step.

The states at t_k are then, starting from t_0,

$$\begin{aligned}
x_0 &= x_0 & & \\
x_1 &= A_0 x_0 + B_0 u_0 & &= A_0 x_0 + B_0 u_0 \\
x_2 &= A_1 x_1 + B_1 u_1 & &= A_1 A_0 x_0 + A_1 B_0 u_0 + B_1 u_1 \\
x_3 &= A_2 x_2 + B_2 u_2 & &= A_2 A_1 A_0 x_0 + A_2 A_1 B_0 u_0 + A_2 B_1 u_1 + B_2 u_2 \\
&\vdots & &\vdots
\end{aligned}$$

The batch programming approach for the QP MPC then has

$$S_x = \begin{bmatrix} I \\ A_0 \\ A_0 A_1 \\ A_0 A_1 A_2 \\ \vdots \\ \prod_{k=0}^{N} A_k \end{bmatrix}$$

$$S_u = \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots & 0 \\ B_0 & 0 & 0 & 0 & \cdots & 0 \\ A_1 B_0 & B_1 & 0 & 0 & \cdots & 0 \\ A_2 A_1 B_0 & A_2 B_1 & B_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \\ \left(\prod_{k=1}^{N-1} A_k\right) B_0 & \left(\prod_{k=2}^{N-1} A_k\right) B_1 & \left(\prod_{k=3}^{N-1} A_k\right) B_2 & \cdots & & B_N \end{bmatrix}$$

$$\bar{Q} = \text{blkdiag}(Q_0, Q_1, \cdots, Q_{N-1}, P)$$

$$\bar{R} = \text{blkdiag}(R_0, R_1, \cdots, R_{N-1})$$